
Assessing User Modeling Bias in Large Language Models for Adaptive Learning

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Abstract

Large Language Models (LLMs) are increasingly used in education for adaptive learning. However, we must ensure that the scaffolding is solely based on competence and abilities rather than demographic characteristics. Despite growing concern regarding bias, there is limited research on auditing LLM user modeling. In this study, we introduce Interrogative Bias Detection to detect demographic bias in LLMs. We examine the question output of a user modeling a persona (Global North student vs. Global South student) at specific coding levels (beginner, intermediate, high intermediate). We use human evaluators to assess the technical depth and cognitive ability inference of the questions for a small subset ($n = 60$). We then use LLMs-as-a-judge after validating their correlation with human evaluators on $N = 188$. We discover that, within the scope of our pilot study, Amazon Nova Lite exhibits no statistically significant bias ($p > 0.05$) in modeling student performance between demographics. We validate that the LLM associates technical proficiency and learner autonomy regardless of the student’s background. We release our code on GitHub: <https://github.com/tiaL-ops/interrogativebias>.

1 Introduction

LLMs have been widely used in education, especially in adaptive learning, to improve student learning outcomes (Goslen et al., 2025). Because of their ability to adapt to users, Kamruzzaman and Kim (2025) raised concerns regarding how LLMs favor Western Europeans and simulate Eastern Europeans, Latin Americans, and Africans negatively.

To address this, Chen et al. (2024) developed a dashboard to visually examine how LLama (a white-box model) models its users. However, there is limited research on auditing black-box models.

Our approach, *Interrogative Bias Detection*, proposes examining the questions an LLM expects students to ask as a window into its internal user model.

Our research questions:

- Can we develop interpretable probes to detect demographic bias in Amazon Nova Lite student modeling?
- Does Amazon Nova Lite exhibit differential modeling of student competence based on geographic or cultural background?
- Can GPT-4o be used as an evaluator for adaptive learning content?

2 Methodology

We developed the Interrogative Bias Dataset to probe how an LLM models a student’s thought process.

Preprint.

Experimental Design

We introduce two personas (American (Global North representative) and Malagasy (Global South representative)) across three coding levels: Beginner (*b*), Intermediate (*i*), and High Intermediate (*hi*). These were distributed across 64 unique contexts. Contexts were manually generated based on a Computer Science curriculum. They included contexts ranging from project initiation to production, as well as social inquiries. We avoided pattern recognition by randomly sampling situational contexts within each coding level while maintaining level-specific context constraints (*b*: 48, *i*: 64, *hi*: 76).

Procedure

Inspired by the simulation outline from Aher et al. (2023), we prompted Amazon Nova Lite to generate questions that these personas would ask.

Evaluation

We asked three human evaluators, acting as blind evaluators, to assess bias and stereotypes in the questions. No evaluator was exposed to the persona, level, or context of each question.

Rubric Evaluation: (1) **Tech Usage Difference** (0–3), (2) **Cognitive Independence** (0–3), (3) **Technical Depth** (0–3), (4) **Question Intent Distribution:** Foundational learning, product implementation, product optimization, DSA, career/social. (4) **Depth vs. Independence Correlation** (Spearman ρ).

3 Results

- **GPT-4o as an evaluator:** GPT-4o demonstrated performance similar to that of human evaluators. We observed no statistically significant difference ($p > 0.05$) at $N = 64$ (32 per group).
- **Adaptation to coding level:** Amazon Nova Lite showed a strong correlation between question depth and independence ($p = 0.019$) at $N = 376$ (188 per group).
- **Demographic user modeling bias:** We observed no statistically significant difference between American and Malagasy students in user modeling outcomes ($p > 0.05$) for $N = 376$ (188 per group).

These results indicate that Amazon Nova Lite generation is aligned with competence rather than demographic background.

4 Limitations and Conclusions

- This study is a pilot with a small sample size; further examinations should be conducted with larger sample sizes and additional demographic personas.
- Amazon Nova Lite shows no statistically significant difference in modeling student performance between demographics in this pilot study and holds potential for adaptive education environments requiring low-cost models.

References

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